FURTHER MECHANICS / CENTRE OF MASS

Centre of Mass of Laminae by Integration

Mark scheme 10 questions · 115 marks

EXAM BOARDS Edexcel FM2 9FM0/4C · AQA 7367/3M · OCR A Y543 · OCR MEI Mech Major Y421

Questions

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	Total	**115**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes. A grey line gives the rule where the name alone doesn't.

ft	follow through Correct work from the student's own earlier wrong value earns the mark.	**AG**	answer given The answer is printed in the question, so every step to it must be shown.
cao	correct answer only No follow-through from an earlier error.	**cso**	correct solution only No error anywhere in the part, not just in its answer.
oe	or equivalent	**awrt**	answers which round to
soi	seen or implied Earned if written, or if later working shows it was used.	**isw**	ignore subsequent working A correct answer, once reached, isn't spoiled by later working.
SC	special case Credit, as the note says, for work outside the main method.	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1(a)	$6x - x^2 = x \Rightarrow x = 0,\ x = 5$	**B1**	
	$\int_0^5 (6x - x^2)\,\mathrm{d}x = \left[3x^2 - \frac{1}{3}x^3\right]_0^5$ $\left[= \frac{100}{3}\right]$	**M1**	Integrates $6x - x^2$, or $5x - x^2$, with at least one power increased, and substitutes their limits.
	$A = \frac{100}{3} - \frac{25}{2} = \frac{125}{6}$	**A1**	
	Triangle: area $\frac{25}{2}$, centre of mass $\left(\frac{10}{3}, \frac{5}{3}\right)$	**B1**	All three values, soi. Where x is subtracted inside each integral instead, for forming $5x - x^2$ and $\frac{1}{2}\left((6x - x^2)^2 - x^2\right)$, soi.
	"$\frac{125}{6}$" $\bar{x} = \int_0^5 (6x^2 - x^3)\,\mathrm{d}x - \frac{25}{2} \times \frac{10}{3}$ $= \frac{375}{4} - \frac{125}{3} \Rightarrow \bar{x} = \ldots$	**M1**	Integrates $x(6x - x^2)$, at least one power increased, and forms their area $\times\, \bar{x}$ = this moment minus their triangle's; or integrates $x(5x - x^2)$ directly.
	$\bar{x} = \frac{5}{2}$	**A1**	
	"$\frac{125}{6}$" $\bar{y} = \frac{1}{2}\int_0^5 (6x - x^2)^2\,\mathrm{d}x - \frac{25}{2} \times \frac{5}{3}$ $= 125 - \frac{125}{6} \Rightarrow \bar{y} = \ldots$	**M1**	Integrates $\frac{1}{2}(6x - x^2)^2$, at least one power increased, and forms their area $\times\, \bar{y}$ = this moment minus their triangle's; or integrates $\frac{1}{2}\left((6x - x^2)^2 - x^2\right)$ directly.
	$\bar{y} = 5$	**A1**	
		[8]	
1(b)	Takes PG vertical: $\tan\phi = \dfrac{3 - \text{"}\frac{5}{2}\text{"}}{9 - \text{"}5\text{"}}$	**M1**	Takes PG as vertical and uses the coordinate differences of P and their G in a tangent, gradient or vector method for an angle. Condone the ratio inverted.
	$\tan\phi = \frac{1}{8}$	**A1ft**	ft their G. Accept ϕ awrt $7.1°$, or a correct angle between PG and the edge.
	$45° - \arctan\frac{1}{8} = 38°$	**A1**	
		[3]	
	Total for Question 1	**11**	

Question	Scheme	Marks	Notes
2(a)	$A = \int_0^{\pi/2}(2+3\cos x)\,\mathrm{d}x$ $= [2x+3\sin x]_0^{\pi/2} = \ldots$	**M1**	Integrates $2+3\cos x$, at least one term correct, and substitutes the limits 0 and $\frac{\pi}{2}$.
	$A = \pi + 3$	**A1**	
	$\int x\cos x\,\mathrm{d}x = x\sin x - \int \sin x\,\mathrm{d}x$	**M1**	Applies integration by parts to $x\cos x$ the right way round, reaching the form $px\sin x \pm q\int \sin x\,\mathrm{d}x$. Condone sign and coefficient slips.
	$\int_0^{\pi/2} x(2+3\cos x)\,\mathrm{d}x$ $= \left[x^2 + 3x\sin x + 3\cos x\right]_0^{\pi/2}$	**A1**	Correct integral of $x(2+3\cos x)$, oe.
	$\bar{x} = \dfrac{\frac{\pi^2}{4} + \frac{3\pi}{2} - 3}{\pi + 3}$ $= \dfrac{\pi^2 + 6\pi - 12}{4(\pi+3)}$	**A1 AG**	Correct limits substituted into the correct integral, divided by $\pi+3$, leading without error to the printed result; combined steps accepted.
		[5]	
2(b)	$\int_0^{\pi/2}\left(4 + 12\cos x + 9\cos^2 x\right)\mathrm{d}x$ $= \int_0^{\pi/2}\left(\frac{17}{2} + 12\cos x + \frac{9}{2}\cos 2x\right)\mathrm{d}x$	**M1**	Expands $(2+3\cos x)^2$ to three terms, uses $\cos^2 x = \frac{1}{2}(1 \pm \cos 2x)$ and integrates to the form $px + q\sin x + r\sin 2x$. Condone coefficient slips.
	$= \left[\frac{17x}{2} + 12\sin x + \frac{9}{4}\sin 2x\right]_0^{\pi/2}$	**A1**	Correct integral, oe.
	$\bar{y} = \dfrac{1}{2(\pi+3)}\left(\dfrac{17\pi}{4} + 12\right) = \dfrac{17\pi + 48}{8(\pi+3)}$	**A1**	Any exact equivalent.
	$\bar{y} - 2 = \dfrac{17\pi + 48 - 16(\pi+3)}{8(\pi+3)}$	**M1**	Compares their exact $\bar{y}$ with 2 without decimals, by forming $\bar{y} - 2$ over a common denominator or comparing numerators over the same denominator.
	$= \dfrac{\pi}{8(\pi+3)} > 0$ so the centre of mass lies above $y = 2$	**A1 AG**	From the correct exact $\bar{y}$, reaches $\frac{\pi}{8(\pi+3)} > 0$ or $17\pi + 48 > 16\pi + 48$ oe without error, and concludes in words. A0 if decimals are used.
		[5]	
	Total for Question 2	**10**	

Question	Scheme	Marks	Notes
3(a)	$\int_0^3 \left(6 - \frac{x^3}{9}\right) \mathrm{d}x = \left[6x - \frac{x^4}{36}\right]_0^3$	**M1**	Integrates $6 - \frac{x^3}{9}$, or $6 - \frac{x^3}{9} - x$ for L directly, increasing at least one power by 1, and uses the limits 0 and 3.
	$= \frac{63}{4}$ $\Rightarrow \mathcal{A} = \frac{63}{4} - \frac{9}{2} = \frac{45}{4}$	**A1**	Area of L, oe.
	$\frac{1}{2}\int_0^3 \left(6 - \frac{x^3}{9}\right)^2 \mathrm{d}x$ $= \frac{1}{2}\left[36x - \frac{x^4}{3} + \frac{x^7}{567}\right]_0^3$	**M1**	Uses $\frac{1}{2}\int y^2 \,\mathrm{d}x$ with limits 0 and 3, or a direct equivalent for L, squaring to three terms and increasing at least one power by 1.
	$= \frac{297}{7}$ $\left[\Rightarrow y = \frac{132}{49}\right]$	**A1**	Either value, oe, or the moment of L directly.
	$\frac{45}{4}\bar{y} = \frac{63}{4} \times \frac{132}{49} - \frac{9}{2} \times 1$ $\Rightarrow \bar{y} = \frac{118}{35}$	**A1 AG**	Correct substituted moment equation for L, the triangle's moment subtracted or by direct integration, leading without error to $\bar{y} = \frac{118}{35}$.
		[5]	
3(b)	$\int_0^3 x\left(6 - \frac{x^3}{9}\right) \mathrm{d}x = \left[3x^2 - \frac{x^5}{45}\right]_0^3$	**M1**	Integrates xy for the curve, or $x\left(6 - \frac{x^3}{9} - x\right)$ for L, with limits 0 and 3, increasing at least one power by 1.
	$= \frac{108}{5}$ $\left[\Rightarrow x = \frac{48}{35}\right]$	**A1**	Either value, oe, or the moment of L directly.
	$\frac{45}{4}\bar{x} = \text{“}\frac{108}{5}\text{”} - \frac{9}{2} \times 2$ $\Rightarrow \bar{x} = \frac{28}{25}$	**A1**	oe, such as 1.12.
	$G\left(\text{“}\frac{28}{25}\text{”}, \frac{118}{35}\right)$ vertically below B: $\tan\phi = \dfrac{\frac{118}{35} - 3}{3 - \text{“}\frac{28}{25}\text{”}}$	**M1**	Takes G vertically below B and uses the offsets of G from B in a ratio for an angle at B. Condone the ratio inverted. Vectors or the cosine rule for angle OBG accepted.
	$\tan\phi = \frac{65}{329}$ $[\Rightarrow \phi = 11.2°]$	**A1ft**	ft their $\bar{x}$. Accept the angle, or a correct ratio for another angle between BG and an axis direction.
	Angle $= 45° + \phi = 56.2°$	**A1**	awrt 56.2°.
		[6]	
	Total for Question 3	**11**	

Question	Scheme	Marks	Notes
4(a)	$\int_0^{\ln 3} 2e^x \,\mathrm{d}x = [2e^x]_0^{\ln 3} = \ldots \ [= 4]$	**M1**	Integrates $2e^x$, or $2e^x - 1$, to the form $ke^x \ (-x)$ and substitutes both limits 0 and $\ln 3$.
	$\mathcal{A} = 4 - \ln 3$	**A1**	Exact, or awrt 2.90.
	$\int_0^{\ln 3} \frac{1}{2}(2e^x)^2 \,\mathrm{d}x = \left[e^{2x}\right]_0^{\ln 3} = \ldots$	**M1**	Integrates $\frac{1}{2}y^2$ for the curve, or $\frac{1}{2}(y^2-1)$ over R, to the form $ke^{2x} \ (+cx)$ and uses the limits. Needs the $\frac{1}{2}$.
	Moment 8, or height 2, for the region under the curve	**A1**	Either value; for integration over R directly, moment $8 - \frac{1}{2}\ln 3$.
	$(4 - \ln 3)\bar{y} = 4(2) - \ln 3\left(\frac{1}{2}\right)$ $\Rightarrow \bar{y} = \frac{16 - \ln 3}{8 - 2\ln 3}$	**A1 AG**	Correct substituted moments equation, rectangle subtracted or integral over R, leading without error to the printed result; combined steps accepted.
		[5]	
4(b)	$\int_0^{\ln 3} 2xe^x \,\mathrm{d}x$ $= [2xe^x]_0^{\ln 3} - \int_0^{\ln 3} 2e^x \,\mathrm{d}x$	**M1**	Integrates by parts with $u = x$, reaching the form $pxe^x - \int qe^x \,\mathrm{d}x$. Condone sign and coefficient slips.
	$= 6\ln 3 - 4$	**A1**	oe, or $x = \frac{3\ln 3 - 2}{2}$ for the region under the curve; for integration over R directly, $6\ln 3 - 4 - \frac{1}{2}(\ln 3)^2$.
	$(4 - \ln 3)\bar{x} = 6\ln 3 - 4 - \ln 3\left(\frac{1}{2}\ln 3\right)$ $\Rightarrow \bar{x} = \frac{12\ln 3 - 8 - (\ln 3)^2}{8 - 2\ln 3}$	**A1**	Any exact equivalent.
	G above and right of D, as $\bar{y} - 2 > 0$: $\tan(180° - \theta) = \frac{\text{“}\bar{x}\text{”}}{\bar{y} - 2}$	**M1**	Uses the offsets $\bar{x}$ and $\bar{y} - 2$ of G from D in the tangent of an angle at D with the vertical or horizontal. Condone the ratio inverted.
	$\tan(180° - \theta) = \frac{12\ln 3 - 8 - (\ln 3)^2}{3\ln 3} \ [= 1.21]$	**A1ft**	ft their $\bar{x}$. Exact, or awrt 1.21; awrt 0.829 for the tangent of the angle with the horizontal.
	$\theta = 129.7°$	**A1**	awrt 129.7°; A0 for the acute angle.
		[6]	
	Total for Question 4	**11**	

Question	Scheme	Marks	Notes
5(a)	$A = \int_0^3 \frac{k}{x+1}\,\mathrm{d}x = k\left[\ln(x+1)\right]_0^3$	**M1**	Integrates to the form $c\ln(x+1)$ and substitutes the limits 0 and 3.
	$A = k\ln 4$	**A1**	oe, such as $2k\ln 2$.
	$\int_0^3 xy\,\mathrm{d}x = k\int_0^3 \left(1 - \frac{1}{x+1}\right)\mathrm{d}x$ $= k\left[x - \ln(x+1)\right]_0^3$	**M1**	Splits $\frac{x}{x+1}$ as $1 - \frac{1}{x+1}$, or uses $u = x+1$ or parts, and integrates to an x (or u) term and a logarithm. Condone sign slips.
	$= k(3 - \ln 4)$	**A1**	
	$\bar{x} = \frac{k(3-\ln 4)}{k\ln 4} = \frac{3-\ln 4}{\ln 4}$	**A1**	oe exact, such as $\frac{3}{\ln 4} - 1$.
		[5]	
5(b)	$\bar{y} = \frac{1}{2\text{“}k\ln 4\text{”}}\int_0^3 \frac{k^2}{(x+1)^2}\,\mathrm{d}x$ $= \frac{k}{2\ln 4}\left[-\frac{1}{x+1}\right]_0^3 = \ldots$	**M1**	Uses $\frac{1}{2A}\int y^2\,\mathrm{d}x$ with their area of the form ck, and integrates $(x+1)^{-2}$ to the form $c(x+1)^{-1}$. Needs the factor $\frac{1}{2}$.
	$\bar{y} = \frac{3k}{8\ln 4}$	**A1**	oe
	$\text{“}\frac{3k}{8\ln 4}\text{”} = \text{“}\frac{3-\ln 4}{\ln 4}\text{”} \Rightarrow k = \ldots$	**M1**	Sets their $\bar{y}$, of the form ck, equal to their $\bar{x}$ and solves for k.
	$k = \frac{8(3-\ln 4)}{3}$	**A1**	oe exact, such as $8 - \frac{16}{3}\ln 2$.
	$A = \text{“}k\text{”}\ln 4 = \frac{8(3-\ln 4)\ln 4}{3}$	**A1ft**	ft their k in their area from (a) of the form ck; exact.
		[5]	
	Total for Question 5	**10**	

Question	Scheme	Marks	Notes
6(a)	$x = 4 - \frac{y^2}{4}$ $A = \int_k^4 \left(4 - \frac{y^2}{4}\right) \mathrm{d}y$ $= \left[4y - \frac{y^3}{12}\right]_k^4$	**M1**	Integrates their x in terms of y with limits k and 4, the power of at least one term increased by 1; oe with vertical strips.
	$A = \frac{32}{3} - 4k + \frac{k^3}{12}$	**A1**	oe, such as $\frac{k^3-48k+128}{12}$.
	$A\bar{y} = \int_k^4 y\left(4 - \frac{y^2}{4}\right) \mathrm{d}y$ $= \left[2y^2 - \frac{y^4}{16}\right]_k^4$	**M1**	Integrates y times their x in terms of y with limits k and 4, the power of at least one term increased by 1; oe with vertical strips.
	$A\bar{y} = 16 - 2k^2 + \frac{k^4}{16}$	**A1**	oe, such as $\frac{(k^2-16)^2}{16}$.
	$\bar{y} = \frac{3(k^2-16)^2}{4(k^3-48k+128)}$ $= \frac{3(k-4)^2(k+4)^2}{4(k-4)^2(k+8)}$	**dM1**	Divides their moment by their area and attempts to factorise both to cancel a common factor.
	$\bar{y} = \frac{3(k+4)^2}{4(k+8)}$	**A1**	cso. Any equivalent with no common factor left between numerator and denominator.
		[6]	
6(b)	$\frac{3(k+4)^2}{4(k+8)} = 3 \Rightarrow (k+4)^2 = 4(k+8)$ $\Rightarrow k^2 + 4k - 16 = 0 \Rightarrow k = \ldots$	**M1**	Sets their $\bar{y}$ from (a), a function of k, equal to 3, reaches a three-term quadratic in k, or a quartic from which they extract it, and solves it by any method.
	$k = -2 \pm 2\sqrt{5}$	**A1**	Correct exact positive root, oe such as $2(\sqrt{5}-1)$; the negative root may also be shown.
	$-2 - 2\sqrt{5} < 0$, rejected as $0 < k < 4$ $k = 2(\sqrt{5} - 1)$	**E1**	Justifies rejecting $-2 - 2\sqrt{5}$ because it is negative, or outside $0 < k < 4$. Where $k = 4$ arises, it must also be rejected, as $k < 4$.
		[3]	
	Total for Question 6	**9**	

Question	Scheme	Marks	Notes
7(a)	$\int_0^{2\ln 2} 4e^{-x/2}\,\mathrm{d}x = \left[-8e^{-x/2}\right]_0^{2\ln 2} = \ldots \ [= 4]$	**M1**	Integrates to the form $ke^{-x/2}$ and substitutes both limits, 0 and $2\ln 2$. Integrating $4e^{-x/2} - 1$ directly also scores.
	$A = 4 - 2\ln 2$	**A1**	Any exact equivalent.
	$\frac{1}{2}\int_0^{2\ln 2} 16e^{-x}\,\mathrm{d}x = 8\left[-e^{-x}\right]_0^{2\ln 2} = \ldots$	**M1**	Integrates $\frac{1}{2}y^2$, squared to the form ke^{-x}, and substitutes both limits. The $\frac{1}{2}$ may be missing; condone a slip in the constant. $\frac{1}{2}\left(16e^{-x} - 1\right)$ directly also scores.
	$= 6 \Rightarrow y_C = \frac{3}{2}$	**A1**	Moment 6 or $y_C = \frac{3}{2}$; or moment $6 - \ln 2$ for the lamina directly.
	$(4 - 2\ln 2)\bar{y} = 4\left(\frac{3}{2}\right) - (2\ln 2)\left(\frac{1}{2}\right)$ $\bar{y} = \frac{6-\ln 2}{4-2\ln 2}$	**A1**	Any exact equivalent.
		[5]	
7(b)	$\int_0^{2\ln 2} 4xe^{-x/2}\,\mathrm{d}x = \left[-8xe^{-x/2}\right]_0^{2\ln 2} + \int_0^{2\ln 2} 8e^{-x/2}\,\mathrm{d}x$	**M1**	Uses parts with $u = x$, reaching the form $pxe^{-x/2} + \int qe^{-x/2}\,\mathrm{d}x$. Condone sign and coefficient slips.
	$= 8 - 8\ln 2$	**A1**	Moment $8 - 8\ln 2$ oe, or $x_C = 2 - 2\ln 2$, or implied by a correct direct moment.
	$(4 - 2\ln 2)x_0 =$ "$8 - 8\ln 2$" $- (2\ln 2)(\ln 2)$	**M1**	Subtracts their rectangle's area times its x-coordinate from their moment under the curve, or integrates the $-x$ term of $x(4e^{-x/2} - 1)$ directly. Condone sign slips.
	$x_0 = \frac{8-8\ln 2-2(\ln 2)^2}{4-2\ln 2}$	**A1**	oe. Implied by the moment $8 - 8\ln 2 - 2(\ln 2)^2$ used correctly in the final equation.
	$A_T = \frac{3}{2}, \quad x_T = 2\ln 2 + 1$	**B1**	Both values, or the triangle's area and moment $\frac{3}{2} + 3\ln 2$ by integration.
	$\left(4 - 2\ln 2 + \frac{3}{2}\right)\bar{x}$ $=$ "$8 - 8\ln 2 - 2(\ln 2)^2$" $+ \frac{3}{2}$"$(2\ln 2 + 1)$"	**M1**	Adds their original lamina's moment to their triangle's moment (area times x-coordinate, or by integration), and divides by the sum of the areas.

Question 7 continued

Question	Scheme	Marks	Notes
	$\bar{x} = \frac{19-10\ln 2-4(\ln 2)^2}{11-4\ln 2}$	**A1**	Any exact equivalent.
		[7]	
	Total for Question 7	**12**	

Question	Scheme	Marks	Notes
8(a)	$\mathrm{d}x = 5\cos t\,\mathrm{d}t,\ y = 4\cos t$ $x = \frac{5}{2}, 5 \Rightarrow t = \frac{\pi}{6}, \frac{\pi}{2}$	**M1**	Expresses y and $\mathrm{d}x$ in terms of t in $\int y\,\mathrm{d}x$ and changes both limits to t, or reverts to x and uses the x limits. Condone a coefficient slip.
	$\mathcal{A} = 20\int_{\pi/6}^{\pi/2}\cos^2 t\,\mathrm{d}t$	**A1**	
	$\mathcal{A} = 10\left[t + \frac{1}{2}\sin 2t\right]_{\pi/6}^{\pi/2}$ $= 10\left[\frac{\pi}{2} - \frac{\pi}{6} - \frac{\sqrt{3}}{4}\right]$	**dM1**	Uses $\cos^2 t = \frac{1}{2}(1 \pm \cos 2t)$, integrates to the form $pt + q\sin 2t$ and substitutes their limits in t. Condone sign and coefficient slips.
	$\mathcal{A} = \frac{10\pi}{3} - \frac{5\sqrt{3}}{2} = \frac{5}{6}\left(4\pi - 3\sqrt{3}\right)$	**A1 AG**	Correct exact substituted limits leading without error to the printed area. Not from decimals.
		[4]	
8(b)	$\int_{5/2}^{5} xy\,\mathrm{d}x = \frac{4}{5}\int_{5/2}^{5} x(25-x^2)^{1/2}\,\mathrm{d}x$ $= \left[-\frac{4}{15}(25-x^2)^{3/2}\right]_{5/2}^{5} = \ldots$	**M1**	Integrates xy to the form $k(25-x^2)^{3/2}$, or $k\cos^3 t$ with $x = 5\sin t$, and substitutes the limits. Condone sign and coefficient slips.
	$\int_{5/2}^{5} xy\,\mathrm{d}x = \frac{25\sqrt{3}}{2}$	**A1**	
	$\int_{5/2}^{5} \frac{1}{2}y^2\,\mathrm{d}x = 8\int_{5/2}^{5}\left(1 - \frac{x^2}{25}\right)\mathrm{d}x$ $= 8\left[x - \frac{x^3}{75}\right]_{5/2}^{5} = \ldots$	**M1**	Writes y^2 in terms of x, integrates with at least one power increased by 1 and substitutes the limits. Condone a missing $\frac{1}{2}$.
	$\int_{5/2}^{5} \frac{1}{2}y^2\,\mathrm{d}x = \frac{25}{3}$	**A1**	
	$\bar{x} = \dfrac{25\sqrt{3}}{2\mathcal{A}} = \dfrac{15\sqrt{3}}{4\pi - 3\sqrt{3}}$	**A1**	Any exact form.
	$\bar{y} = \dfrac{25}{3\mathcal{A}} = \dfrac{10}{4\pi - 3\sqrt{3}}$	**A1**	Any exact form.
		[6]	
8(c)	G below A: $\tan\theta = \dfrac{\text{“}\bar{y}\text{”}}{5 - \text{“}\bar{x}\text{”}}$	**M1**	Takes AG vertical and forms $\tan\theta$ as the vertical over the horizontal offset of G from A, or an equivalent trigonometric ratio, with their $\bar{x}$, $\bar{y}$ from (b). Condone the ratio inverted.
	$\tan\theta = \dfrac{1}{2\pi - 3\sqrt{3}}$ $[= 0.920]$	**A1ft**	ft their $\bar{x}$ and $\bar{y}$; exact or decimal.

Question 8 continued

Question	Scheme	Marks	Notes
	$\theta = 42.6°$	**A1**	awrt $42.6°$.
		[3]	
	Total for Question 8	**13**	

Question	Scheme	Marks	Notes
9(a)	$\int_0^{\frac{\sqrt{3}a}{2}} \sqrt{a^2-y^2}\,\mathrm{d}y,\ y=a\sin t$ $= a^2\int_0^{\pi/3}\cos^2 t\,\mathrm{d}t$	**M1**	Uses a trigonometric substitution, such as $y=a\sin t$ or $x=a\cos t$, in an integral of $\sqrt{a^2-y^2}$ or $\sqrt{a^2-x^2}$, changing the limits or reverting. Condone a slip in the differential.
	$= \frac{a^2}{2}\left[t+\frac{\sin 2t}{2}\right]_0^{\pi/3}$	**dM1**	Uses $\cos^2 t = \frac{1}{2}(1\pm\cos 2t)$ oe, integrates to the form $pt+q\sin 2t$ and substitutes the limits. Condone coefficient slips.
	$= a^2\left(\frac{\pi}{6}+\frac{\sqrt{3}}{8}\right)$	**A1**	Or the correct value of the equivalent integral in another method.
	Rectangle: $\frac{a}{2}\times\frac{\sqrt{3}a}{2}=\frac{\sqrt{3}a^2}{4}$	**B1**	Or $\frac{\sqrt{3}a^2}{2}$ for the whole rectangle. Implied by a correct method needing no rectangle.
	$M = 2\sigma\left(a^2\left(\frac{\pi}{6}+\frac{\sqrt{3}}{8}\right)-\frac{\sqrt{3}a^2}{4}\right)$ $= \frac{\sigma a^2(4\pi-3\sqrt{3})}{12}$	**A1**	Any exact equivalent including σ, such as $\sigma a^2\left(\frac{\pi}{3}-\frac{\sqrt{3}}{4}\right)$.
		[5]	
9(b)	Plate symmetric about the x-axis, so $\bar{y}=0$	**E1**	States the symmetry about the x-axis, or the bisector of angle AOB. Not for $\bar{y}=0$ alone.
	Rectangle: $\frac{\sqrt{3}a^2}{4}\times\frac{a}{4}=\frac{\sqrt{3}a^3}{16}$	**B1**	Or $\frac{\sqrt{3}a^3}{8}$ for the whole rectangle, with or without σ. Implied by a correct method needing no rectangle.
	$\frac{1}{2}\int_0^{\frac{\sqrt{3}a}{2}}(a^2-y^2)\,\mathrm{d}y$ $= \frac{1}{2}\left[a^2y-\frac{y^3}{3}\right]_0^{\frac{\sqrt{3}a}{2}}$	**M1**	Integrates $\frac{1}{2}x^2$ with $x^2=a^2-y^2$ with respect to y, power of a term increased by 1, or $2x\sqrt{a^2-x^2}$ with respect to x to the form $k(a^2-x^2)^{3/2}$.
	$M\bar{x} = 2\sigma\left(\frac{3\sqrt{3}a^3}{16}-\frac{\sqrt{3}a^3}{16}\right)=\frac{\sqrt{3}\sigma a^3}{4}$	**A1**	Or $\frac{\sqrt{3}a^3}{4}$ as a moment of area, used consistently.
	$\bar{x} = \frac{\sqrt{3}\sigma a^3}{4}\div\frac{\sigma a^2(4\pi-3\sqrt{3})}{12}=\frac{3\sqrt{3}a}{4\pi-3\sqrt{3}}$	**A1**	Any exact equivalent, such as $\frac{9a}{4\sqrt{3}\pi-9}$.
		[5]	
	Total for Question 9	**10**	

Question	Scheme	Marks	Notes
10(a)	$\mathcal{A} = \int_0^2 kx^2 \,\mathrm{d}x = k\left[\frac{x^3}{3}\right]_0^2$	**M1**	Integrates kx^2, power increased by 1, and uses the limits 0 and 2.
	$\mathcal{A} = \frac{8k}{3}$	**A1**	
	$\int_0^2 xy \,\mathrm{d}x = \int_0^2 kx^3 \,\mathrm{d}x$ $= k\left[\frac{x^4}{4}\right]_0^2$	**M1**	Integrates $x \times kx^2$, or an equivalent strip integral, power increased by 1, with the limits used.
	$\int_0^2 xy \,\mathrm{d}x = 4k$	**A1**	
	$\bar{x} = 4k \div \frac{8k}{3} = \frac{3}{2}$	**A1**	From a correct area and moment.
	$\frac{1}{2}\int_0^2 y^2 \,\mathrm{d}x = \frac{1}{2}\int_0^2 k^2x^4 \,\mathrm{d}x$ $= \frac{k^2}{2}\left[\frac{x^5}{5}\right]_0^2$	**M1**	Integrates $(kx^2)^2$, power increased by 1, with the limits used. The $\frac{1}{2}$ may be missing; condone k not squared.
	$\frac{1}{2}\int_0^2 y^2 \,\mathrm{d}x = \frac{16k^2}{5}$	**A1**	Or $\int_0^2 y^2 \,\mathrm{d}x = \frac{32k^2}{5}$ when later divided by $2\mathcal{A}$.
	$\bar{y} = \frac{16k^2}{5} \div \frac{8k}{3} = \frac{6k}{5}$	**A1**	
		[8]	
10(b)	Curved piece: area $\frac{8k}{3}$, centre $\left(2 + \frac{6k}{5}, \frac{3}{2}\right)$	**B1ft**	ft their (a) as $(2 + \text{“}\bar{y}\text{”}, \text{“}\bar{x}\text{”})$.
	$\left(4 + \frac{8k}{3}\right)\bar{x} = 4 \times 1$ $+ \frac{8k}{3}\left(\text{“}2 + \frac{6k}{5}\text{”}\right)$	**M1**	Forms total area $\times \bar{x}$ = sum of moments, with both pieces, area $\frac{8k}{3}$ for the curved piece and their x-coordinate of its centre.
	$\bar{x} = \frac{12k^2+20k+15}{10k+15}$	**A1**	oe; need not be simplified.
	$\left(4 + \frac{8k}{3}\right)\bar{y} = 4 \times 1 + \frac{8k}{3} \times \text{“}\frac{3}{2}\text{”}$	**M1**	Forms the corresponding equation for $\bar{y}$, with their y-coordinate of the curved piece's centre.
	$\bar{y} = \frac{3(k+1)}{2k+3}$	**A1**	oe; need not be simplified.
		[5]	
10(c)	$W = T_A + T_D = 3T_D$	**B1**	oe, such as $T_A = \frac{2}{3}W$, or $T_A = 2T_D$ used in a moments equation about G. Implied by $\bar{x} = \frac{4}{3}$ from a correct method.
	Moments about A: $W(2 - \bar{x}) = 2T_D$	**M1**	Takes moments about any point, with the weight acting at $x = \bar{x}$ and every tension that has a moment there. Condone sign slips.

Question 10 continued

Question	Scheme	Marks	Notes
	$\bar{x} = \frac{4}{3}$	**A1**	
	$\frac{12k^2+20k+15}{10k+15} = \text{“}\frac{4}{3}\text{”}$ $\Rightarrow 36k^2 + 20k - 15 = 0 \Rightarrow k = \ldots$	**M1**	Equates their $\bar{x}$ from (b), a quadratic over a linear expression in k, to their value, and solves the resulting three-term quadratic by any means.
	$k = \frac{4\sqrt{10}-5}{18}$	**A1**	Any exact equivalent, with the negative root rejected or not given.
		[5]	
	Total for Question 10	**18**	